



Stargazing Through Time: How Ancient Cultures Used the Night Sky

【BBC精选：观星寻路 古文明如何用夜空导航、计时与生存】

Summary: The video explores how ancient cultures across the world utilized the night sky for navigation, timekeeping, and cultural practices. It highlights Aboriginal Australians reading the Milky Way to track emu migrations, Polynesian navigators memorizing star sequences for ocean voyages, and megalithic structures like Stonehenge and New Grange aligned with solstices. The segment also touches on Peru's solar calendar towers and concludes with concerns about modern light pollution obscuring celestial knowledge and disrupting wildlife.

摘要： 本片揭示了古文明如何运用夜空进行导航、计时和文化活动。澳大利亚原住民通过银河暗斑判断鸸鹋产卵期，波利尼西亚人依靠记忆星序横渡太平洋，巨石阵与新格朗日古墓则精准对齐冬至阳光。视频还展示了秘鲁的太阳历塔群，最后指出光污染正使人类失去星空，并危及依赖星辰导航的候鸟。





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🕒 Estimated Reading Time: 7 min.

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▶ It's night time in the Australian Outback and it's dark, very dark.

这是澳大利亚内陆的夜晚，漆黑一片，黑得不能再黑。

▶ But if you look up, the sky is ablaze with light.

但如果你抬头，天空却#燃烧般明亮#。

▶ We're looking towards the heart of the Milky Way galaxy,

我们正望向#银河#的中心，

▶ a vast pinwheel of about 100 billion stars.

那是一座由约一千亿颗星星组成的巨大旋涡。

▶ Can you see the emu?

你能看到那只鸸鹋吗？

▶ Yeah, that's the one.

是的，就是那一个。

▶ It's made of dark clouds of dust and gas, obscuring the lighter parts of the sky.

它由黑暗的尘埃与气体云构成，遮住了天空中较#明亮#的部分。

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▶ There are several legends about how the emu was banished to the heavens.

关于这只鸸鹋如何被放逐到天上，有好几个传说。

▶ But in the Aboriginal nation of Wyragerie,

但在怀拉杰里的#原住民#部族中，

▶ her position in the sky signals the time of year best for collecting emu eggs.

她在天空中的位置预示着一年中最适合收集鸸鹋蛋的时节。

▶ For most of human history, understanding the sky was not the preserve of scientists,

在人类的大部分历史中，理解天空并不是科学家的专属，

▶ but a precious tool for our survival, for building cultures, religions and societies,

而是我们#生存#所需的#珍贵#工具，用来建立文化、宗教和社会，

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▶ and in some places it still is.

在某些地方至今依然如此。

▶ This is Orion the Hunter.

这是猎户座。

▶ You're probably familiar with his belt and sword, but if you're close to the equator, 你大概熟悉他的腰带和#剑#, 但如果你接近#赤道#,

▶ it looks a lot more like a canoe.

它看起来更像一只#独木舟#。

▶ A thousand years ago, the Polynesians were the most accomplished navigators in the world.

一千年前, 波利尼西亚人是世界上最#高超#的航海家。

▶ Up to the 11th century, they explored over 1,000 tiny islands scattered across the Pacific Ocean.

直到11世纪, 他们探索了太平洋上分布的上千个#零散#小岛。

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▶ One of the ways they did this was by using an ingenious memory technique.

他们做到这一点的方法之一是使用一种#巧妙#的记忆技巧。

▶ With a stone canoe built on the beach,

在沙滩上建造一艘石头#独木舟#,

▶ a training navigator would face the sea and sit in there night after night,

一名正在训练的航海士会面朝大海, 夜复一夜地坐在里面,

▶ memorizing the rising and falling positions of the stars.

背诵星辰的升起与落下的位置。

▶ When the crew was ready to depart, they would head for a star near the horizon,

当船员准备好#启航#时, 他们会朝着接近#地平线#的一颗星星前进,

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▶ switching to a new one once the first either rose or set as it moved across the sky.

当第一颗星升起或落下时，就切换到下一颗星继续导航。

⏮ Each sailing route required a specific sequence of stars, like a password or a musical motif.

每条航线都需要一套特定的星星#序列#，就像一个密码或一个音乐#主题#。

⏮ The celestial navigation, using the sun, moon and stars to work out precisely where you are in the world,

这种#天文# #导航# —— 利用太阳、月亮和星辰来精确判断你在世界上的位置，

⏮ was essential to your appearance, if errors too.

对他们的生存至关重要，尽管也难免会出错。

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⏮ But what if it wasn't a navigation tool you were after, but a way of judging time?

但如果你追求的不是#导航#工具，而是判断时间的方法呢？

⏮ Some have suggested that Stonehenge in England was built

有人认为，英国的巨石阵建造出来

⏮ as an astronomical computer of great sophistication, capable of predicting eclipses.

是为了成为一个高度#复杂#的#天文#计算机，可以预测日食。

⏮ This can't be proven, it's most likely to be a precise way of determining the winter solstice,

这无法被证明，更可能是用来精确确定冬至，

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⏮ perfect for timing observances around the shortest day of the year.

以便安排一年中最短白天的祭祀活动。

⏮ Stonehenge is solstitially aligned, meaning if you're approaching the entrance from the northeast,

巨石阵与冬至对齐，这意味着如果你从#东北#方向的#入口#进入，

▶ the sunset and midwinter will cut the monument neatly in half.

冬至时的#日落#会把这个#纪念碑#整齐地一分为二。

▶ New Grange in Ireland doesn't even better job.

爱尔兰的新格兰奇甚至做得更好。

▶ The 19-meter passage runs the length of a long barrow leading to a central chamber.

一条长达19#米#的通道贯穿整个长形#古坟#，通向中央墓室。

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▶ In the mornings around the winter solstice,

在冬至前后的早晨，

▶ the sun shines directly in through a crack above the doorway.

阳光会直接透过#门口#上方的一道缝隙照入。

▶ New Grange was a tomb, so for the people that built it,

新格兰奇是一座#坟墓#，因此对于建造它的人来说，

▶ ancestors, death, the sun and seasonal renewal were probably linked.

祖先、死亡、太阳与季节的轮回大概是联系在一起的。

▶ Perhaps it was the job of the ancestors to turn the sun around at midwinter, guaranteeing the return of spring.

也许在他们看来，祖先的职责就是在冬至时扭转太阳，确保春天的归来。

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▶ One of the most stunning examples of an ancient astronomical tool is at Chankillo in Peru.

秘鲁冲基约的一处遗迹是古代#天文#工具中最#惊艳#的例子之一。

▶ A line of 13 towers provided a detailed solar calendar to someone viewing it from especially Bill's doorway.

一排13座塔楼，从比尔的#门口#观看，可以形成一个精确的太阳#历法#。

▶ People may have gathered from miles around to celebrate certain days of the year.

人们可能会从数英里外赶来庆祝一年中特定的日子。

▶ But with the development of complex technologies such as GPS,

但随着GPS等复杂技术的发展，

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▶ most humans no longer rely on the sun, moon and stars to travel or to organize their lives.

大多数人类不再依赖太阳、月亮和星辰来旅行或安排生活。

▶ Just as well because light pollution is increasing globally between 2 and 10% year on year,

这也算幸运，因为全球光污染每年以2%到10%的速度增加，

▶ only one in five people in Western Europe and America can now see the Milky Way.

如今在西欧和美国，只有五分之一的人还能看到#银河#。

▶ But birds need the stars, including the North Star, to navigate.

但是鸟类需要依靠星星，包括北极星，来#导航#。

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▶ If they can't see them due to light pollution from cities, they run into trouble.

如果因为城市的光污染看不见星星，它们就会遇到麻烦。

▶ Denying animals' access to the stars is just one of the many ways in which light pollution impacts nature,

剥夺动物观星的能力只是光污染影响自然的众多方式之一，

▶ but that's another story.

但那是另一个故事了。

▶ Reliance on the sky as a guiding light is not uniquely human.

把天空作为指引之光的这种#依赖#并非人类独有。

⏮ But perhaps our use of the sky to find food, to organize our lives, to socialize,

但也许我们利用天空寻找食物、安排生活、进行#社交#，

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⏮ to pray, was not just something we did before better technology became available.

以及祈祷，并不仅仅是因为当时缺乏更好的技术。

⏮ Perhaps it was what made us who we are.

或许，这正是造就我们成为人类的原因。

